

Kaleb Coberly

📍 Bellingham, WA, USA | ✉ kaleb.coberly@gmail.com | 🌐 kalebcoberly.com

Summary

Backend software engineer and data engineer/analyst with a background in research, analytics, and data science, plus training and project experience in machine learning. Builds, tests, and deploys tools, research software, analytics, and data pipelines with an emphasis on reliability, automation, reproducibility, and maintainability. Experienced with Python, R, SQL, ETL, automated testing, Docker, and CI/CD, and supporting researchers and developers through debugging, documentation, training, and developer tooling. Curious, interdisciplinary engineer committed to continued learning across software, data, emerging technologies, and the domains they serve.

Skills

- *Programming & Scripting*: Python, R, Bash, SQL, NoSQL, C++.
- *DevOps & Tooling*: Git, Make, Docker, GitHub Actions, Jenkins, CI/CD, Python packaging, Kubernetes (K8s), QE and testing frameworks.
- *Data Engineering*: ETL, APIs, data cleaning/validation/versioning, RDBMS, DB design, MySQL, MariaDB, MongoDB, Elasticsearch, HTML/XML/JSON/YAML/CSV.
- *ML Engineering*: ML pipelines, feature engineering, performance evaluation, model validation.
- *Analysis & Visualization*: EDA, Jupyter Notebook, RStudio, R Markdown, Mermaid, Excel, Kibana, Tableau.
- *Documentation & Support*: Sphinx, reStructuredText, user training, contributor onboarding, issue triage.
- *Platforms & Environments*: MacOS, Windows, Linux, HPC (Slurm, UGE), Anaconda, GitHub, Bitbucket/Stash, VS Code, JIRA, Asana.

Experience

Research Engineer, IHME – Central Computation (GBD), Nov 2022–Oct 2024

- www.healthdata.org/research-analysis/gbd
- Developed and maintained internal Python and dev tools for data modeling, versioning, access, and dev workflows.
- Contributed to a large in-house Python ecosystem: features, bugfixes, major test coverage increases, and documentation.

- Maintained CI/CD pipelines with Jenkins builds and integrated testing, monitored build health across ~100 packages.
- Debugged issues across DBs, Python APIs/CLIs, Docker, k8s, and HPC clusters.
- Directly supported research teams: Investigated failed model runs, data access issues, and tooling gaps.
- Analyzed system usage and tool performance via ElasticSearch; ran ad-hoc reports and built automated reports.

Data Analyst, IHME – Pandemics Team, Nov 2021–Dec 2022

- Owned multiple data pipelines for COVID-19 production; reduced key runtime by 65%.
- Refactored and extended vaccine and booster modeling pipeline.
- Built diagnostics, with automated reporting for data QA/QC/QE; inspected and cleaned data.
- Led early UGE-to-Slurm transition for production jobs.
- Built ETL pipeline in R of national data into crosswalked, versioned output.
- Produced executive reports for world leaders, running ETL into R Markdown for PDFs; adapted to changing model outputs and feature requests:
 - www.healthdata.org/research-analysis/diseases-injuries/covid/policy-briefings

Volunteer Software Engineer, Crickets and Comb, Nov 2024–Present

- cricketsandcomb.org
- Built and maintain reusable GitHub Actions CI/CD pipelines and dev flows with automated testing and QE for local + Docker development and open-source Python package releases, github.com/crickets-and-comb/shared.
 - Maintain Python version and package support window.
 - Led AI agent in migrating package suite and central tools from `pytype` to `mypy`: github.com/orgs/crickets-and-comb/projects/3.
 - Migrated package suite to Python 3.14 compatibility: github.com/crickets-and-comb/shared/issues/108.
- Onboard developers by using documentation, presentations, and hands-on support of testing conventions, CI steps, and Git Flow.
- Automated part of Bellingham Food Bank's delivery operations, reducing several staff-hours with each weekly use, github.com/crickets-and-comb/bfb_delivery:
 - Designed and implemented an ETL CLI integrated with staff workflows and 3rd-party API.
 - Achieved high test coverage, by line count, problem space, and abstraction level.
 - Led incremental design, rollout, and user training.
 - Wrote and deployed technical and non-technical documentation.
 - Maintain ongoing user support and dependency management.
 - Released as open-source project; lead collaborative development at all skill levels.

Volunteer ML/Software Engineer, Cascade STEAM, April 2025–April 2025

- cascadesteam.org

- Leading the Citizen Science group's development of computer-vision ETL tool for stormwater monitoring field observation datasheets:
 - github.com/crickets-and-comb/stormwater_monitoring_datasheet_extraction

Education

B.S., Database Management and Data Analytics – Western Governors University

B.A., Critical Studies and Pedagogy – The Evergreen State College

Certifications

- Quantum Computing (Qiskit & Python) – Udemy, Apr 2025
- Mathematics for ML & Data Science – DeepLearning.AI, Jan 2025
- Oracle Database SQL Certified Associate – Oracle, Jul 2020
- CompTIA A+, Network+, Project+ – CompTIA, 2019 / 2020

References

Available upon request.